



Research paper

Determining the absolute abundance of dinoflagellate cysts in recent marine sediments II: Further tests of the *Lycopodium* marker-grain methodKenneth Neil Mertens^{a,*}, Andrea Michelle Price^b, Vera Pospelova^b^a Research Unit Palaeontology, Ghent University, Krijgslaan 281 S8, 9000 Gent, Belgium^b School of Earth and Ocean Sciences, University of Victoria, OEASB A405, P.O. Box 1700 STN CSC, Victoria, BC, Canada V8W 2Y2

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ABSTRACT

Lycopodium clavatum tablets are commonly added as a spike to determine dinoflagellate cyst concentrations in sediments. In this study we investigate the effects of different processing techniques on dinoflagellate cyst concentrations using well-mixed sediment samples from Saanich Inlet, British Columbia, Canada. At the onset of any dinoflagellate cyst investigation, we suggest following the recommendations of Maher (1981) to experimentally adjust the sample size to obtain a ratio close to ~2 of dinoflagellate cysts counted to *Lycopodium* spores counted, in order to obtain reproducible concentrations. Results further show that both oven-drying at ~45 °C and freeze-drying are viable, non-destructive techniques yielding reproducible results. Use of warm HCl (40–60 °C) for a short time (30 min) is harmless, whereas treatment with warm HF (40–60 °C) affects the reproducibility of the concentrations. Pre-sieving can result in loss of cysts and/or spike but this can be easily monitored by checking the residue. Perforated metal sieves show more consistent results than the Nitex nylon meshes. The use of 30 second sonication does not affect the reproducibility, and is advised to remove amorphous organic matter. Adding the *Lycopodium* spike at the end of preparation yields consistently lower concentrations, which were usually not reproducible, suggesting noticeable losses of *Lycopodium* spores during processing if the *Lycopodium* spores are added at the beginning. This method can be considered a viable alternative, but the discrepancy should be taken into account.

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1. Introduction

Precise quantification of dinoflagellate cyst abundance is extremely important to make reliable estimates of productivity, important for both environmental studies (e.g. Matsuoka et al., 2003; Dale, 2008) and paleoceanographical studies (e.g. Marret et al., 2001; Pospelova et al., 2006; Mertens et al., 2009a; Penaud et al., 2010). Dinoflagellate cysts are extracted using palynological methods, which can be highly variable between different laboratories, and often involve the use of hydrochloric (HCl), hydrofluoric (HF) and/or alternative acids at different temperatures (e.g. Wood et al., 1996; Riding and Kyffin-Hughes, 2004, 2009, 2011; Riding et al., 2007).

In general, dinoflagellate cyst concentrations are calculated by adding a known amount of exotic markers or a “spike” to each sample according to the method described by Stockmarr (1971). The marker commonly used is *Lycopodium clavatum* Linnaeus (Stag's Horn Clubmoss or Ground Pine). A recent interlaboratory study by Mertens et al. (2009b) showed that variation in processing techniques between laboratories resulted in significant differences in calculated concentrations using the *Lycopodium* marker-grain method. Similar investigations

have shown uncertainties associated with this method (Lignum et al., 2008).

In this paper we further optimize the *Lycopodium* marker-grain method by studying the effects of different processing techniques on surface sediment samples collected from three stations in Saanich Inlet, British Columbia, Canada. The effects studied here are (1) drying-techniques, (2) use of warm acids instead of cold (room temperature) acids, (3) use of sieve type (metal or nylon), (4) use of pre-sieving, (5) use of sonication, and (6) an alternative method of adding *Lycopodium* tablets at the end of the preparation instead of at the beginning. We subsequently make recommendations for further standardization.

2. Material and methods

2.1. Palynological processing and statistical analysis

Surface sediment samples were collected using a Van Veen grab during a cruise in Saanich Inlet on May 19, 2009 at three stations, respectively station A (= S2) (48°33.148' N, 123°31.969' W, water depth 215 m), station B in Patricia Bay (= PB3) (48°39.005' N, 123°29.027' W, water depth 98 m) and station C (= S3) (48°35.475' N, 123°30.211' W, water depth 228 m) (Fig. 1). The lithology is a very fine, dark-colored, organic rich mud.

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The samples were processed using the following methodology unless otherwise noted and follows recommendations by Mertens et al. (2009b). This method is illustrated in Fig. 2, with an indication of the steps that were modified in the respective tests. The sediment was homogenized by mixing thoroughly, and subsampled into 50 mL polypropylene centrifuge tubes (~1 g was used for station A, ~2 g for station B and ~3 g for station C). Freeze-drying was used unless otherwise noted. No pre-sieving (i.e. sieving before use of acids) was done unless otherwise noted. An average of 7–9 mL of cold (room temperature) 10% HCl was added for 20–35 min. A fixed number of *Lycopodium clavatum* tablets of batch #177745 ($X=18,584$ per tablet, $\sigma=\pm 1853$) or batch #483216 ($X=18,583$ per tablet, $\sigma=\pm 1708$), were simultaneously added with the HCl to each sample, unless otherwise noted. The *Lycopodium* spore tablets used are produced and distributed by the Department of Quaternary Geology, University of Lund, Sweden (<http://www.geol.lu.se/kvg/eng/>). After 20–35 min, the residue was concentrated by centrifuging the tubes at 3000 rpm for 6 min, and was then rinsed with distilled water. Approximately 7–15 mL of cold 48% HF was then added for 48 h. The residue was then concentrated by centrifuging the tubes at 3000 rpm for 6 min, treated with cold 10% HCl for 10–15 min, centrifuged again, and rinsed twice with distilled water. Except for the three samples processed in test 6, no sonication was used during any of the processing to overcome any potential loss of cysts, previously experienced by Mertens et al. (2009b) when long sonication (>1 min) was used. No heavy liquid was used. The residues were sieved using stackable metal 15 μm sieves, made of nickel and are mounted in a stainless steel frame. They are manufactured (Stork Veco International, Inc.) by the electroforming technique which is a deposition process that results in very uniform hole sizes and a smooth surface. The sieve holes are punched in and not interlaced. The hole tolerance is $\pm 2 \mu\text{m}$. Slides were mounted on a heated metal plate (65 °C) using a pipette and stewed using a toothpick. The mounting medium was glycerin jelly. To overcome observer bias, all counting was done by the first author using a Nikon Eclipse

80i transmitting light microscope at 600 magnification. Dinoflagellate cyst specimens were counted only when they comprised at least half of a cyst. A minimum of 300 dinoflagellate cysts were counted (Mertens et al., 2009b). Indeterminate dinoflagellate cysts were grouped as Indeterminate spp. Raw counts are available as Supplementary data to this article. Absolute abundances of dinoflagellate cysts were calculated following the equation by Benninghoff (1962).

$$c = \frac{d_c \times L_t \times t}{L_c \times w}$$

where

c	concentration = number of dinoflagellate cysts/g dried sediment
d_c	number of counted dinoflagellate cysts
L_t	number of <i>Lycopodium</i> spores/tablet
t	number of tablets added to the sample
L_c	number of counted <i>Lycopodium</i> spores
w	weight of dried sediment (g)

Maier (1981) devised an algorithm to calculate confidence limits on microfossil concentrations. As in Mertens et al. (2009b), a slight correction to this algorithm was made, since the current study used sediment weight instead of sediment volume. The confidence limits calculated based on this algorithm have a 0.95 probability ($Z=1.95$). It should be noted that these confidence limits are similar to the total error on concentrations proposed by Stockmarr (1971); (cf. Mertens et al., 2009b). These confidence limits can then be used in a statistical test to check whether microfossil concentrations are the same in two different samples (Maier, 1981, p. 180–184). Dinoflagellate cyst concentrations from each test are here compared to three replicate samples using Maier's test of reproducibility. A sample is statistically similar to the replicate when it falls within

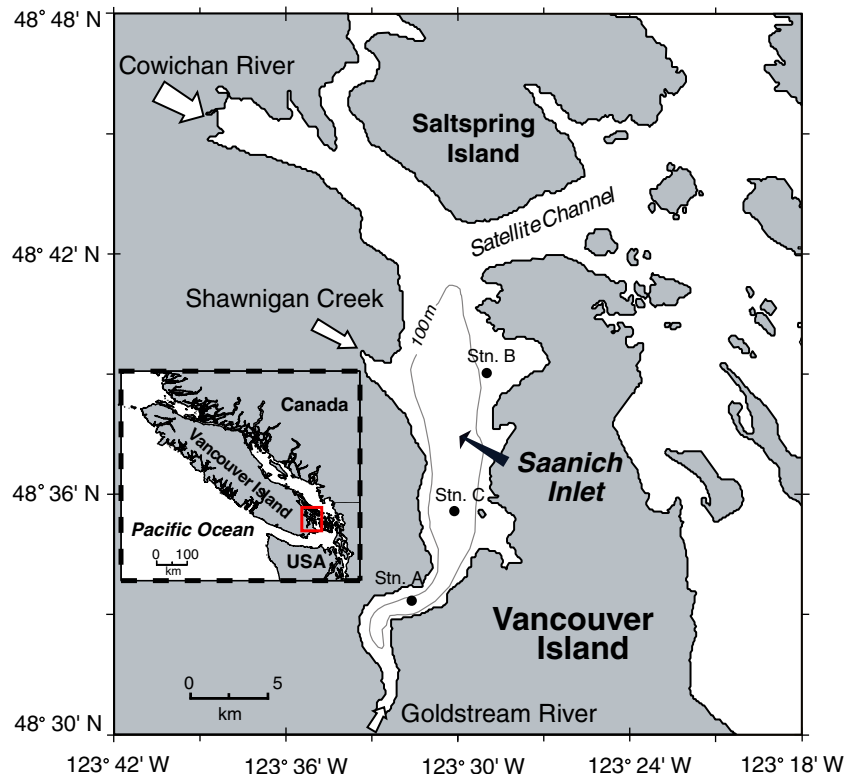


Fig. 1. Map of Saanich Inlet, British Columbia (Canada) showing locations of sampling stations A, B and C.

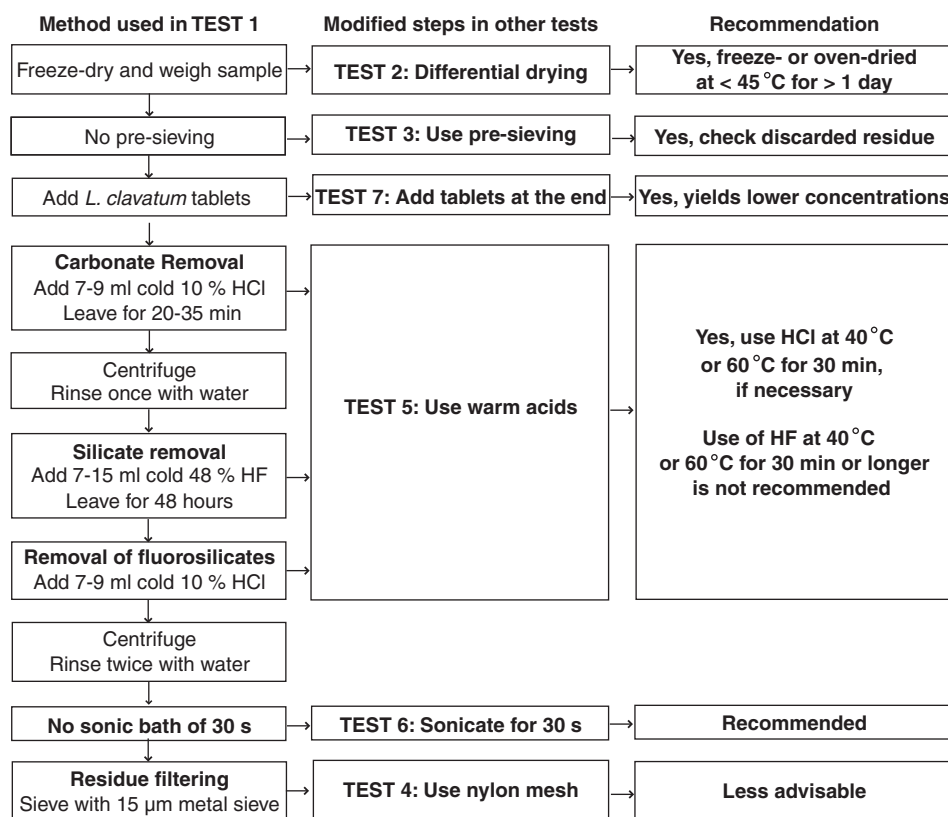


Fig. 2. Flow-chart showing the method used in test number 1 in the first column. Second column shows the steps modified in the other tests. Third column shows the recommendations in regard to these tests.

the confidence limits and is deemed acceptable, while a sample is rejected when it does not fall in the confidence limits.

2.1.1. Test 1: the freeze-dried samples are reproducible

As in Mertens et al. (2009b) we first tested if samples that were processed in a similar manner yielded reproducible results. Three replicas of samples from each station (A, B, and C) were freeze-dried and processed.

2.1.2. Test 2: differential drying

To estimate the weight loss of the wet samples, 20 samples of different weights up to 50 g were dried at ~45 °C and weighed daily until the samples had completely dried.

To test the effect of drying, one subsample from each station was processed using the following drying methods: wet (without drying), freeze-drying, and drying in an oven at ~45 °C for 1 day, 1 week, and 1 month. Also, one freeze-dried set of subsamples was left in the dark at room temperature for 1 year.

2.1.3. Test 3: effect of pre-sieving

To test the hypothesis whether pre-sieving causes losses (hypothesis 9 in Mertens et al., 2009b), three samples were processed with the use of pre-sieving at 15 µm before the use of acids and three samples with pre-sieving at both 15 µm and 120 µm before the use of acids.

2.1.4. Test 4: nylon mesh instead of metal sieve

To test the hypothesis whether sieving with a 15 µm nylon mesh causes any losses (hypothesis 7 in Mertens et al., 2009b), three samples were processed with a nylon mesh.

2.1.5. Test 5: the effect of cold vs. warm acid treatment (HCl and HF)

Dale (1976) mentioned that hot rinses with HCl after the HF treatment were particularly harmful to recent peridinoïd cysts, and

Mertens et al. (2009b) mentioned explicit deterioration of the cysts when hot HF treatments were used at temperatures between 42 and 80 °C. To further explore the possible effect of warm acids one set of three samples was processed using warm HCl for half an hour (one set at 40 °C and one set at 60 °C), two sets using warm HF for half an hour (one set at 40 °C and one set at 60 °C), two sets using warm HF for 12 h at 60 °C (one set was freeze-dried and another oven-dried) and one last set using both acids (each acid for half an hour at 60 °C).

2.1.6. Test 6: 30 second sonication

To test whether any cysts were lost by using short sonication times (30 s), we processed one set of samples using the method that is currently used at the Paleoenvironmental Laboratory, University of Victoria, Canada (Pospelova et al., 2005; Krepakevich and Pospelova, 2011). In this method, samples were oven dried for 4 days, pre-sieved with a 15 µm and 120 µm nylon sieve, gently sonicated for 30 s using a Branson™ sonic bath and collected on a 15 µm nylon sieve.

2.1.7. Test 7: adding *Lycopodium* tablets at end of preparation

As recommended by Mertens et al. (2009b) as an alternative to adding *Lycopodium* tablets at the beginning of processing, we tested adding *Lycopodium* tablets at the end of preparation.

2.2. Geochemical methods

To determine total organic carbon (TOC) content of the samples, one sediment sample from stations A, B and C was analyzed to determine total carbon (TC) and calcium carbonate (CaCO₃) content. Analysis was done at the Pacific Centre for Isotopic and Geochemical Research (PCIGR), University of British Columbia, Canada.

3. Results

3.1. Relative abundance of dinoflagellate cysts

The assemblages encountered at the three stations are similar to the ones described in the Saanich Inlet sediment-trap study by Price and Pospelova (2011). Thirty-one taxa were determined and counted. The assemblage is dominated by heterotrophic species: *Brigantedinium cariacense*, *Brigantedinium simplex*, *Brigantedinium* spp., cyst of *Polykrikos kofoidii*, cyst of *Polykrikos schwartzii*, cyst of *Protoperidinium americanum*, cyst type L, *Dubridinium* spp., *Echinidinium aculeatum*, *Echinidinium* cf. *delicatum*, *Echinidinium granulatum*, *Echinidinium transparentum*, *Protoperidinium* spp., *Quinquecupis concreta*, round brown cysts, *Selenopemphix quanta*, *Selenopemphix undulata*, spiny brown cysts (SBC), *Trinovantedinium variabile*, *Votadinium calvum*, *Votadinium spinosum* and rare occurrences of cyst of *Protoperidinium nudum* and cyst type A. Autotrophic cysts are rare and consist of cyst of *Pentapharsodinium dalei*, *Operculodinium centrocarpum*, *Spiniferites bentorii*, *Spiniferites ramosus*, *Spiniferites* spp., and very rare occurrences of cyst of *Alexandrium* spp., *Spiniferites elongatus* and *Spiniferites membranaceus*. A small percentage of the dinoflagellate cyst assemblage remained indeterminate. The average relative abundances for the three stations are summarized in Table 1. The cyst of *Biecheleria* sp. was also encountered in the three stations, but this species was excluded from the relative abundances and absolute abundances because of its large size range (~10–30 µm), where some of them can get easily lost during sieving (see Price and Pospelova (2011) for a description). Reworked dinoflagellate cysts were extremely rare and only 2 specimens were encountered amongst the 18,189 dinoflagellate cysts counted during this study and were not included in the dinoflagellate cyst counts.

Quantitative and qualitative disparities between assemblages at each site may be due to the different processing methods. It is obvious that aggressive agents could destroy the more sensitive cysts. To

check this dependence of preservation on methodology, it is necessary to group the present species according to their resistance to degradation. It is assumed that both mechanical and chemical degradation have similar effects on an assemblage. The grouping shown below is similar to the grouping described by Zonneveld et al. (2001). Cysts not referred to by these authors were added to a particular group based on the assumption that comparable morphology (e.g. wall thickness, resistance of structures against folding) is indicative of similar resistance to decay.

Extremely sensitive cysts: *Brigantedinium cariacense*, *Brigantedinium simplex*, *Brigantedinium* spp., cyst of *Alexandrium* spp., cyst of *Polykrikos kofoidii*, cyst of *Polykrikos schwartzii*, cyst of *Protoperidinium americanum*, cyst of *Protoperidinium nudum*, cyst type A, cyst type L, *Dubridinium* spp., *Echinidinium aculeatum*, *Echinidinium* cf. *delicatum*, *Echinidinium granulatum*, *Echinidinium transparentum*, *Protoperidinium* spp., round brown cysts (RBC), SBC, *Selenopemphix quanta*, *Selenopemphix undulata* and *Trinovantedinium variabile*.

Moderately sensitive cysts: *Operculodinium centrocarpum*, *Quinquecupis concreta*, *Spiniferites bentorii*, *Spiniferites elongatus*, *Spiniferites membranaceus*, *Spiniferites ramosus*, *Spiniferites* spp., *Votadinium calvum* and *Votadinium spinosum*.

Resistant cysts: cyst of *Pentapharsodinium dalei*.

For station A, when averaging out all counts regardless of the processing method used, there was an average of 91.3% extremely sensitive cysts, 8.4% moderately sensitive cysts and 0.2% resistant cysts (Table 2). The standard deviations of the relative abundances observed in the grouping based on cyst preservation are always lower than 1.7%. Similarly, for station B, there was an average of 77.5% extremely sensitive cysts, 21.9% moderately sensitive cysts and 0.6% resistant cysts (Table 3). The standard deviations of the relative abundances observed in the grouping based on cyst preservation are always lower than 6.9%. For station C, there was an average of 91.4% extremely sensitive cysts, 8.5% moderately sensitive cysts and 0.2% resistant cysts (Table 3). The standard deviations of the relative abundances observed in the grouping based on cyst preservation are always lower than 3.4%. The relatively small standard deviations for the three stations suggest that changes in the relative abundance counts are caused by in situ variations rather than by differences in methodology. Changes in the relative abundances thus show no relation to differences in processing methodology. Also, no evident changes in visual appearance of the cysts could be related to changes in methodology.

3.2. Absolute abundances of dinoflagellate cysts

The reproducibility of the calculated concentrations in the following paragraphs is tested using Maher's method (1981) by comparison to the concentrations of the three replicas. The results are shown in Tables 2–4 and are summarized in Table 5 and below.

3.2.1. Test 1: the freeze-dried samples are reproducible

The dinoflagellate cyst concentration in the three freeze-dried replicas is reproducible for station A and for station B (Table 5). The freeze-dried replicas for station C were not reproducible (Table 5), and also show relatively high differences in relative abundances (i.e. up to 15.8% for RBC).

3.2.2. Test 2: differential drying

The results of oven drying 20 samples from each station showed that, depending on the weight of the sample, it took between 1 and 11 days to dry the sample completely at 45 °C. However, for samples with a dry weight of less than 3 g, drying was already completed after 24–48 h. The average sample water loss was very similar for all stations, and for station A was 87.9% (stdev = 0.42), for station B 71.8% (stdev = 0.76) and for station C 81.8% (stdev = 0.35). These averages are used to correct the wet sample weights of the samples that were processed without drying.

Table 1
Average percentage (%) of the different dinoflagellate cyst taxa in the three samples.

Dinoflagellate cysts	Station A	Station B	Station C
Cyst of <i>Alexandrium</i> spp.	0.00	0.10	0.02
<i>Brigantedinium cariacense</i>	4.52	1.65	3.50
<i>Brigantedinium simplex</i>	0.77	0.17	0.51
<i>Brigantedinium</i> spp.	9.65	5.94	10.21
Round brown cyst	49.40	35.12	48.11
Cyst of <i>Protoperidinium americanum</i>	12.13	10.35	13.45
<i>Protoperidinium</i> spp.	3.41	11.16	4.19
<i>Dubridinium</i> spp.	1.30	1.35	1.31
<i>Echinidinium aculeatum</i>	1.30	0.61	1.13
<i>Trinovantedinium variabile</i>	0.28	0.10	0.21
<i>Echinidinium</i> cf. <i>delicatum</i>	0.70	0.07	0.44
<i>Echinidinium granulatum</i>	1.09	0.71	0.72
<i>Echinidinium transparentum</i>	0.21	0.00	0.03
Cyst type A	0.03	0.13	0.33
Cyst type L	0.39	2.49	0.72
Spiny brown cyst	3.75	4.43	4.65
<i>Operculodinium centrocarpum</i> sensu Wall and Dale (1966)	0.38	4.78	0.76
Cyst of <i>Pentapharsodinium dalei</i>	0.23	0.62	0.18
<i>Polykrikos kofoidii</i>	0.57	0.54	0.39
<i>Polykrikos schwartzii</i>	0.11	0.17	0.08
<i>Quinquecupis concreta</i>	4.34	10.16	5.52
<i>Selenopemphix undulata</i>	1.11	0.59	0.84
Cyst of <i>Protoperidinium nudum</i>	0.02	0.05	0.00
<i>Selenopemphix quanta</i>	0.29	0.71	0.25
<i>Spiniferites bentorii</i>	1.03	1.58	0.28
<i>Spiniferites elongatus</i>	0.02	0.00	0.00
<i>Spiniferites membranaceus</i>	0.00	0.02	0.00
<i>Spiniferites ramosus</i>	0.18	0.49	0.12
<i>Spiniferites</i> spp.	1.63	3.20	1.00
<i>Votadinium calvum</i>	0.33	0.56	0.59
<i>Votadinium spinosum</i>	0.55	0.79	0.16
Indeterminate spp.	0.28	1.38	0.28

Table 2

Results from station A: test number, description of method, calculated concentrations of the samples, results of test reproducibility using [Maher's test \(1981\)](#) when compared to three replicate freeze-dried samples RA1, RA2 and RA3; relative abundance (%) of extremely sensitive cysts, moderately sensitive cysts, and resistant cysts and UVic ID number.

Test #	Method used	Concentration (cysts g ⁻¹ sediment)	Maher's test of reproducibility			Extremely sensitive cysts (%)	Moderately sensitive cysts (%)	Resistant cysts (%)	UVic ID #
			RA1 (136837)	RA2 (98073)	RA3 (121880)				
1	Replica RA1 — freeze-dried	136,837		Accept	Accept	91.33	8.00	0.67	09-105
1	Replica RA2 — freeze-dried	98,073	Accept		Accept	91.85	7.84	0.31	09-106
1	Replica RA3 — freeze-dried	121,880	Accept	Accept		94.61	4.71	0.67	09-107
2a	Oven-dried 1 day	20,941	Reject	Reject	Reject	91.61	7.74	0.65	09-114
2b	Oven-dried 1 week	79,245	Reject	Accept	Accept	91.84	8.16	0.00	09-117
2c	Oven-dried 1 month	137,662	Accept	Accept	Accept	91.67	8.33	0.00	09-120
2d	Wet — corrected weight	118,044	Accept	Accept	Accept	90.03	9.63	0.33	09-148
2e	Freeze-dried — left at room temperature for 1 year	90,629	Accept	Accept	Accept	89.29	10.06	0.65	09-144
3a	Presieved at 15 µm	123,516	Accept	Accept	Accept	91.33	8.67	0.00	11-160
3b	Presieved at 15 µm and 120 µm	142,284	Accept	Accept	Accept	91.69	8.31	0.00	11-163
4	Nylon mesh instead of a metal sieve	82,898	Reject	Accept	Accept	89.94	9.77	0.29	09-138
5a	Warm HCl at 40 °C for 30 min	111,023	Accept	Accept	Accept	91.80	7.87	0.33	09-129
5b	Warm HCl at 60 °C for 30 min	154,088	Accept	Accept	Accept	94.63	5.37	0.00	09-233
5c	Warm HF at 40 °C for 30 min	144,389	Accept	Accept	Accept	91.23	8.12	0.65	09-132
5d	Warm HF at 60 °C for 30 min	157,987	Accept	Accept	Accept	93.36	6.64	0.00	09-236
5e	Warm HF at 60 °C for 12 h, + 14 h at room temperature (freeze-dried)	125,888	Accept	Accept	Accept	90.10	9.90	0.00	09-384
5f	Warm HF at 60 °C for 12 h + 14 h at room temperature (oven-dried)	59,903	Reject	Reject	Reject	89.07	10.93	0.00	09-636
5g	Warm HCl for 30 min and warm HF (60 °C) for 30 min	142,123	Accept	Accept	Accept	91.69	8.31	0.00	09-239
6	Oven-dried 4 days, 15 µm nylon sieve, 30 s sonication	101,176	Accept	Accept	Accept	88.63	11.37	0.00	09-659
7	<i>Lycopodium</i> tablets added at end of preparation	65,646	Reject	Accept	Reject	90.67	9.33	0.00	09-141
	Average	110,712				91.32	8.45	0.23	
	Stdev	35,766				1.61	1.65	0.28	

The concentrations of the samples that were oven-dried for 24 h were distinctly lower and statistically similar to none of the three replicas (Table 5).

The concentrations of the samples that were oven-dried for 1 week were higher than samples that were oven-dried for 24 h, and were statistically similar to two of the three replicas for station

Table 3

Results from station B: test number, description of method, calculated concentrations of the samples, results of test reproducibility using [Maher's test \(1981\)](#) when compared to three replicate freeze-dried samples RB1, RB2 and RB3; relative abundance (%) of extremely sensitive cysts, moderately sensitive cysts, and resistant cysts and UVic ID number.

Test #	Method used	Concentration (cysts g ⁻¹ sediment)	Maher's test of reproducibility			Extremely sensitive cysts (%)	Moderately sensitive cysts (%)	Resistant cysts (%)	UVic ID #
			RB1 (44316)	RB2 (33841)	RB3 (28784)				
1	Replica RB1 — freeze-dried	44,316		Accept	Accept	84.30	13.65	2.05	09-108
1	Replica RB2 — freeze-dried	33,841	Accept		Accept	76.77	23.23	0.00	09-109
1	Replica RB3 — freeze-dried	28,784	Accept	Accept		88.29	11.71	0.00	09-110
2a	Oven-dried 1 day	12,623	Reject	Reject	Reject	86.78	13.22	0.00	09-115
2b	Oven-dried 1 week	33,728	Accept	Accept	Accept	77.40	22.60	0.00	09-118
2c	Oven-dried 1 month	26,769	Reject	Accept	Accept	75.86	23.79	0.34	09-121
2d	Wet — corrected weight	36,008	Accept	Accept	Accept	89.23	10.77	0.00	09-149
2e	Freeze-dried — left at room temperature for 1 year	35,051	Accept	Accept	Accept	67.32	28.43	4.25	09-145
3a	Presieved at 15 µm	41,012	Accept	Accept	Accept	70.95	29.05	0.00	11-161
3b	Presieved at 15 µm and 120 µm	43,577	Accept	Accept	Accept	74.43	25.25	0.33	11-164
4	Nylon mesh instead of a metal sieve	48,156	Accept	Accept	Reject	70.13	29.87	0.00	09-139
5a	Warm HCl at 40 °C for 30 min	41,827	Accept	Accept	Accept	79.31	20.69	0.00	09-130
5b	Warm HCl at 60 °C for 30 min	30,951	Accept	Accept	Accept	79.87	20.13	0.00	09-234
5c	Warm HF at 40 °C for 30 min	21,539	Reject	Reject	Accept	76.11	23.21	0.68	09-133
5d	Warm HF at 60 °C for 30 min	49,536	Accept	Accept	Reject	76.69	23.31	0.00	09-237
5e	Warm HF at 60 °C for 12 h, + 14 h at room temperature (freeze-dried)	31,735	Accept	Accept	Accept	77.18	22.48	0.34	09-385
5f	Warm HF at 60 °C for 12 h + 14 h at room temperature (oven-dried)	18,995	Reject	Reject	Accept	76.47	23.53	0.00	09-637
5g	Warm HCl for 30 min and warm HF (60 °C) for 30 min	172,514	Reject	Reject	Reject	84.64	15.36	0.00	09-240
6	Oven-dried 4 days, 15 µm nylon sieve, 30 s sonication	34,876	Accept	Accept	Accept	62.62	33.44	3.93	09-660
7	<i>Lycopodium</i> tablets added at end of preparation	20,689	Reject	Reject	Accept	76.17	23.49	0.34	09-142
	Average	40,326				77.53	21.86	0.61	
	Stdev	32,647				6.83	6.22	1.28	

Table 4

Results from station C: test number, description of method, calculated concentrations of the samples, results of test reproducibility using [Maher's test \(1981\)](#) when compared to three replicate freeze-dried samples RC1, RC2 and RC3; relative abundance (%) of extremely sensitive cysts, moderately sensitive cysts, and resistant cysts and UVic ID number.

Test #	Method used	Concentration (cysts g ⁻¹ sediment)	Maher's test of reproducibility			Extremely sensitive cysts (%)	Moderately sensitive cysts (%)	Resistant cysts (%)	UVic ID #
			RC1 (33903)	RB2 (88322)	RB3 (62151)				
1	Replica RC1 — freeze-dried	33,903		Reject	Reject	94.67	5.02	0.31	09-111
1	Replica RC2 — freeze-dried	88,322	Reject		Accept	98.73	1.27	0.00	09-112
1	Replica RC3 — freeze-dried	62,151	Reject	Accept		97.99	2.01	0.00	09-113
2a	Oven-dried 1 day	10,561	Reject	Reject	Reject	87.54	11.78	0.67	09-116
2b	Oven-dried 1 week	22,579	Accept	Reject	Reject	91.30	8.70	0.00	09-119
2c	Oven-dried 1 month	21,862	Accept	Reject	Reject	92.00	8.00	0.00	09-122
2d	Wet — corrected weight	35,061	Accept	Reject	Reject	88.75	10.61	0.64	09-150
2e	Freeze-dried — left at room temperature for 1 year	20,489	Reject	Reject	Reject	90.85	8.83	0.32	09-146
3a	Presieved at 15 µm	27,049	Accept	Reject	Reject	89.04	10.96	0.00	11-162
3b	Presieved at 15 µm and 120 µm	36,680	Accept	Reject	Reject	85.38	14.29	0.33	11-165
4	Nylon mesh instead of a metal sieve	33,619	Accept	Reject	Reject	93.31	6.69	0.00	09-140
5a	Warm HCl at 40 °C for 30 min	30,576	Accept	Reject	Reject	95.02	4.98	0.00	09-131
5b	Warm HCl at 60 °C for 30 min	37,141	Accept	Reject	Accept	90.13	9.54	0.33	09-235
5c	Warm HF at 40 °C for 30 min	28,109	Accept	Reject	Reject	94.00	6.00	0.00	09-134
5d	Warm HF at 60 °C for 30 min	29,152	Accept	Reject	Reject	89.37	10.63	0.00	09-238
5e	Warm HF at 60 °C for 12 h, + 14 h at room temperature (freeze-dried)	22,279	Accept	Reject	Reject	91.28	8.72	0.00	09-386
5f	Warm HF at 60 °C for 12 h + 14 h at room temperature (oven-dried)	20,087	Reject	Reject	Reject	90.27	9.73	0.00	09-638
5g	Warm HCl for 30 min and warm HF (60 °C) for 30 min	26,357	Accept	Reject	Reject	88.37	11.63	0.00	09-241
6	Oven-dried 4 days, 15 µm nylon sieve, 30 s sonication	20,089	Reject	Reject	Reject	89.74	10.26	0.00	09-661
7	<i>Lycopodium</i> tablets added at end of preparation	19,960	Reject	Reject	Reject	89.18	9.84	0.98	09-143
	Average	31,301				91.35	8.47	0.18	
	Stdev	17,051				3.40	3.29	0.29	

A, to all of the three replicas for station B, and to one of the three replicas for station C (Table 5).

The concentrations of the samples that were oven-dried for 1 month were also higher than the samples that were oven-dried for 24 h, and were statistically similar to all of the three replicas for station A, to two of the three replicas for station B, and to one of the three replicas for station C (Table 5).

The concentrations of the samples that were not dried (and where their weight was corrected with the above mentioned values) were also statistically similar to all of the three replicas for stations A and B and to one of the three replicas for station C (Table 5).

The concentrations of the freeze-dried samples that were left for 1 year in a cupboard were statistically similar to all of the three replicas for stations A and B and to none of the three replicas for station C (Table 5).

3.2.3. Test 3: effect of pre-sieving

The concentrations of the samples that were pre-sieved at 15 µm and at both 15 and 120 µm prior to acid treatment were statistically similar to all of the three replicas for station A, to all of the three replicas for station B, and to one of the three replicas for station C (Table 5).

3.2.4. Test 4: nylon mesh instead of metal sieve

The concentrations of the samples that were processed using a nylon mesh were statistically similar to two of the three replicas for station A, to two of the three replicas for station B, and to one of the three replicas for station C (Table 5).

3.2.5. Test 5: the effect of cold vs. warm acid treatment (HCl and HF)

The concentrations of the samples that were treated with warm HCl were statistically similar to all of the replicas for stations A and B but not for station C (details in Table 5). The concentrations of the samples that were treated with warm HF were usually not statistically similar to the three replicas for all stations (details in Table 5).

3.2.6. Test 6: 30 second sonication

The concentrations of the samples where 30 second sonication was used were statistically similar to all of the three replicas for station A, to all of the three replicas for station B and to none of the three replicas for station C (Table 5).

3.2.7. Test 7: adding *Lycopodium* tablets at end of preparation

The concentrations of the samples where *Lycopodium* tablets were added at the end of preparation were noticeably lower than the replicas, and were statistically similar to one of the three replicas for station A, to one of the three replicas for station B and to none of the three replicas for station C (Table 5).

3.3. Geochemical results

TOC contents of the samples were relatively high (A = 4.0%, B = 3.3%, C = 3.3%). Calcium carbonate contents were relatively low (A = 1.7%, B = 0.1%, C = 4.3%).

4. Discussion

4.1. Differences in reproducibility between the freeze-dried samples and exclusion of samples from station C

Stations A and B showed reproducible concentrations, while this was not the case for most samples from station C. This is possibly due to the fact that too few *Lycopodium* tablets were added to samples from station C, resulting in a high ratio of dinoflagellate cysts counted to *Lycopodium* spores counted. This sample indeed had less *Lycopodium* spores added to the initial sample than stations A and B, as we used 3 g of this sample, compared to 1 g for station A and 2 g for station B. [Maher \(1981\)](#) suggested an optimal ratio of ~2 for the pollen/marker ratio (in this case dinoflagellate cysts to *Lycopodium* spores), and station C had the highest ratio (an average of 8.45 for the three freeze-dried samples), whereas stations A and B had smaller

Table 5

Summary results of tests 1–7. Number of times accepted or rejected using Maher's test of reproducibility for Stations A, B, and C, where '+' indicates the sample was statistically similar to a replicate and accepted, while '-' indicates the sample was not statistically similar to a replicate and was rejected. *Station C is excluded, see section 4.1 for details.

Test #	Method used	Station A			Station B			*Station C		
		RA1	RA2	RA3	RB1	RB2	RB3	RC1	RC2	RC3
1	Replica R1 - Freeze-dried		+	+		+	+	-	-	
1	Replica R2 - Freeze-dried	+		+	+		+	-		+
1	Replica R3 - Freeze-dried	+	+		+	+		-	+	
2a	Oven-dried 1 day	-	-	-	-	-	-	-	-	-
2b	Oven-dried 1 week	-	+	+	+	+	+	+	-	-
2c	Oven-dried 1 month	+	+	+	+	+	+	+	+	-
2d	Wet - corrected weight	+	+	+	+	+	+	+	-	-
2e	Freeze-dried- left at room temperature for 1 year	+	+	+	+	+	+	-	-	-
3a	Presieved at 15 µm	+	+	+	+	+	+	+	-	-
3b	Presieved at 15 µm and 120 µm	+	+	+	+	+	+	+	-	-
4	Nylon mesh instead of a metal sieve	-	+	+	+	+	-	+	-	-
5a	Warm HCl at 40 °C for 30 min	+	+	+	+	+	+	+	-	-
5b	Warm HCl at 60 °C for 30 min	+	+	+	+	+	+	+	-	+
5c	Warm HF at 40 °C for 30 min	+	+	+	-	-	+	+	-	-
5d	Warm HF at 60 °C for 30 min	+	+	+	+	+	-	+	-	-
5e	Warm HF at 60 °C for 12 hrs, + 14 hrs at room temperature (freeze-dried)	+	+	+	+	+	+	+	-	-
5f	Warm HF at 60 °C for 12 hrs + 14 hrs at room temperature (oven-dried)	-	-	-	-	-	+	-	-	-
5g	Warm HCl for 30 min and warm HF (60 °C) for 30 min	+	+	+	-	-	-	+	-	-
6	Oven-dried 4 days, 15 µm nylon sieve, 30 s sonication	+	+	+	+	+	+	-	-	-
7	<i>Lycopodium</i> tablets added at end of preparation	-	+	-	-	-	+	-	-	-

ratios closer to 2 (respectively 6.23 and 3.75 for the three freeze-dried samples). Indeed, when three samples from station C were reprocessed using a lower weight, so that this ratio was closer to 2 (respectively 1.6, 1.8 and 1.8), samples were reproducible (respectively 33,374, 39,324 and 38,255 cysts g⁻¹). Because these initial samples from station C were not reproducible, we decided to exclude the results of these samples from the discussion of the other tests.

4.2. Effect of different drying techniques

Since the samples were of high organic matter content (TOC between 3.2 and 4.0%, as shown by the geochemical analysis), drying could cause formation of selenite (gypsum, CaSO₄·2H₂O), by reaction of calcium carbonate with sulfuric acid, usually derived from pyrite decay. Although calcium carbonate is in low abundance in the studied samples (between 0.1 and 4.3%), the reaction can take place. The formation of sulfuric acid could affect the extremely sensitive dinoflagellate cysts. The results show that for most samples of stations A and B results were reproducible when samples had been sufficiently dried (freeze-dried, freeze-dried

and left for 1 year, oven-dried for 1 week, oven-dried for 1 month) or were still wet and the weight was corrected, and thus are all useful techniques for drying. Only when samples were dried for 24 h, were samples not reproducible, which was expected as we observed that these samples were still wet after this amount of time.

4.3. Effect of pre-sieving

The results show that most samples from stations A and B results were reproducible when they were either pre-sieved only at 15 µm or pre-sieved at both 15 µm and 120 µm. However, we should note that pre-sieving at 120 µm without any sonication can cause anomalies in concentrations, either by cysts or *Lycopodium* spores being retained on the coarse sieve by clustering with other particles. This can easily be overcome by transferring the material left on the 120 µm sieve to a beaker, briefly sonicating, pre-sieving again and then checking what is left on the sieve. This can be repeated until no cysts and/or *Lycopodium* spores are left on the 120 µm sieve.

4.4. Using a nylon mesh instead of metal sieve

Stations A and B were less reproducible using a nylon mesh when compared to the use of a metal sieve, although concentrations were not consistently lower or higher. However, the samples that were sonicated were also processed using a nylon sieve and were reproducible when compared to the use of a metal sieve (see Section 3.2.6). The data thus suggest possible introduction of error when a nylon mesh is used.

4.5. Effect of using warm acids

Use of warm HCl at 40 °C or 60 °C for 30 min did not affect the reproducibility of the concentrations for stations A and B. Use of warm HF at 40 °C or 60 °C for 30 min or 12 h decreased the reproducibility of the concentrations of stations A and B significantly. Although freeze-dried samples were always reproducible with HF at 60 °C and for 12 h, in contrast to oven-dried samples, the fact that freeze-dried samples that were treated for 30 min with HF at 40 and 60 °C were often not reproducible, suggests that the drying technique does not explain the difference.

4.6. Effect of 30 second sonication

Adding a round of 30 s sonication does not affect the reproducibility of concentrations for stations A and B, and confirms earlier suggestions by Mertens et al. (2009b) that this short sonication does not affect concentrations. In fact, for sediments from coastal settings like Saanich Inlet, sonication is strongly advised, as this significantly removes a large amount of amorphous organic matter, facilitating the identification of dinoflagellate cysts.

4.7. Adding *Lycopodium* tablets at the end of preparation

Results from this alternative method which added *Lycopodium* at the beginning of preparation, were statistically significantly different. Since concentrations derived from this method are consistently lower, and no *Lycopodium* spores can be lost using this method, *Lycopodium* spores must be lost either during sieving or during chemical processing, resulting in higher relative concentrations. As concentrations for samples from stations A and B are respectively 45% and 42% lower than the average of the three freeze-dried samples, this suggests that a similar amount of spores have been lost during processing. It can be suggested that this mainly occurs during mechanical processing (sieving, decanting), as chemical processing only affects the color of *Lycopodium* spores (Sengupta, 1975). These large losses cannot be confirmed however, as only minor losses are observed when observing what passes

through the used sieves. Similarly, Lignum et al. (2008) only suggested losses up to $5.8 \pm 1.2\%$ for 15 μm nylon meshes. At this stage we fail to explain this discrepancy. Interestingly, concentrations are also consistently lower when the volumetric method is used (Mertens et al., 2009b), which possibly makes these methods comparable.

5. Conclusions and recommendations

- To obtain reproducibility of concentrations, we agree with the suggestion of Maher (1981) to adjust sample size to obtain a ratio close to ~2 of dinoflagellate cysts counted to *Lycopodium* spores counted. This could reduce problems with reproducibility significantly.
- Freeze-drying and oven-drying at 45 °C for a sufficient amount of time (at least more than 1 day, less than 1 week) are both viable drying techniques and do not result in damage to cysts. Samples that have been left for a year after being freeze-dried also do not seem to suffer any losses.
- Pre-sieving at 120 μm can result in the loss of cysts or spikes, but this can be easily monitored by checking the residue captured on the 120 μm mesh.
- Metal sieves of the type used in this study show more consistent results than the nylon meshes used here.
- Use of warm HCl at 40 °C or 60 °C for 30 min does not affect reproducibility, whereas use of warm HF at 40 °C or 60 °C for 30 min or longer affects reproducibility significantly and is not advised.
- A round of 30 s sonication using a Branson™ sonic bath does not affect the reproducibility and is advised to remove amorphous organic matter.
- Adding the *Lycopodium* spike at the end of preparation yields consistently lower dinoflagellate cyst concentrations and can lead to non-reproducibility, suggesting losses of *Lycopodium* spores during processing when the *Lycopodium* spores are added at the beginning. It can be considered a viable alternative or the discrepancy should be taken into account.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.revpalbo.2012.06.012>.

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